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OPTIMISE Your BUSINESS SETUP With PoE TECHNOLOGY

PoE switch

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Power over Ethernet (PoE) technology enables a common cable to deliver current to run devices, as well as data packets for connecting these to a network. PoE can be a powerful tool to ease operations and to maintain essential enterprise appliances, and eventually reduce expenses and downtime.

Where PoE can be applied

Businesses can use PoE to run most of their network-based appliances and data loggers, which require reliable, consistent and uninterrupted power as well as network. Sumit Thukral, chief executive officer, Rational Power Systems, says, "PoE is majorly used with network-based devices like Internet Protocol (IP)-based surveillance cameras, Wi-Fi access points and routers, IP phones and so on, where a dedicated data channel and scalable powering is necessary. It is more profitable in large-scale deployment scenarios like large hotels, hospitals or offices, where powering each individual device with separate current source and Ethernet cable is cumbersome and expensive. PoE can be highly beneficial for low-power wireless devices as well."

A spokesperson from Infonet Solutions, a PoE solution provider, adds, "PoE can also be used in industries for various data acquisition devices and programmable controllers. Telecom and Internet service providers are big users of PoE. For instance, it can power server units and provide dedicated network connectivity through a single cable."

Further applications include radio-frequency identification (RFID) and access controls, industrial lighting systems, attendance systems, Voice over IP (VoIP) phones, building automation setups and so on.

A usual setup of a PoE system starts with a PoE network switch that combines power and data input, and channels these through a cable—usually a standard CAT3, CAT5 or CAT6 cable. It then connects to the powered device, that is, the target device or appliance.

Other setups may include devices like PoE injector, which adds power to a data-only source, or PoE splitter that separates power and data channels based on powered device compatibility.

Benefits of PoE

PoE can bring an array of benefits, including expense reduction, ease of maintenance, quicker servicing and reliable connectivity.

Cost optimisation. Thukral mentions, "The major advantage of PoE is that local power is not required for individual devices. Primary cost savings come in the form of reduced excess wiring and cabling. That itself cuts down the installation cost and labour charges significantly."